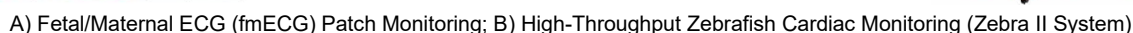


1. Academic Innovation: *Ph.D. in Biomedical Engineering, University of California, Irvine (Advisor: Dr. Hung Cao)*

Technological Innovation: I developed active, non-contact capacitive sensors capable of recording ECG signals through clothing, eliminating the need for wet electrodes. This work, recently published in *IEEE Transactions on Instrumentation and Measurement* (2025), established the foundational high-impedance frontend architectures that I propose to adapt for the Biswas Fellowship's nasal interface. Integrating this sensor technology, I designed a custom **4-layer rigid-flex PCB** with active shielding to minimize interference and implemented **1-D octave convolution neural networks** on edge devices. This optimization reduced computational load by >50% while maintaining >91% fetal heart rate accuracy. Separately, my research on robust fetal ECG extraction algorithms was highlighted by *IEEE Spectrum* in the feature "Smartphone-Based Pregnancy Monitor" (2022).



Global Impact: Building on these results, I established an international partnership with **VinMec Hospital** and **Hanoi University of Science and Technology** in Vietnam. I successfully secured a **~\$235,000 research grant** from the Vingroup Innovation Foundation to support a large-scale trial involving 200 pregnant women, directly addressing the global health equity mission of the Biswas program. Furthermore, I mentored a team of undergraduates at **Vin University** to secure a **\$13,000 VentureWell BME-IDEA grant**, extending the project's educational impact.

Interdisciplinary Collaboration: This work was conducted in partnership with Prof. Xiaolei Xu (Mayo Clinic Research). I deployed to his lab to integrate the system into their workflow, resulting in multiple high-impact publications characterizing cardiac phenotypes.

Commercial Validation: This technology was instrumental in securing a **\$1.6 Million NIH SBIR Phase II grant** for our startup partner, Sensoriis, Inc., validating the commercial and scientific value of my hardware design.

2. Translational Execution: *Senior Electrical Engineer, Vena Vitals, Inc. (YC-backed startup)*

Upon graduation, I transitioned to industry to exactly what it takes to get a device into patient's hands, mastering regulatory compliance, manufacturing, and scale. At Vena Vitals, I lead the electrical engineering team developing a continuous, non-invasive blood pressure monitor.

Rapid Prototyping & Crisis Management: When an external contractor delayed a critical FDA milestone, I took the initiative to architect a superior hardware solution in-house. In just two weeks, I delivered a fully validated prototype that outperformed the external solution, saving the company \$25,000 and de-risking our regulatory timeline.

Clinical Scale & Safety: I currently manage full-cycle hardware development for devices deployed on over 500 patients across multiple clinical sites. I lead our **IEC 60601** safety testing and **EMC/EMI** compliance efforts, ensuring our technology meets the rigorous standards required for FDA 510(k) submission (submitted in Sept, 2025). This experience has instilled a "safety-first" engineering mindset that is critical for the active neuro-stimulation proposed in my Biswas project.



Continuous Non-Invasive Blood Pressure Monitoring System

3. Leadership, Mentorship, & Community Engagement

I believe scientific impact is amplified through mentorship and community building.

Academic Leadership: As the Teaching Assistant for the UCI BioDesign Capstone (100+ students), I served as a technical advisor for 23 student teams, mentoring them on FDA regulatory strategy, technical feasibility, and business modeling. My guidance was frequently cited by students as the pivotal factor in their project success.

Mentoring the Next Generation: I have directly supervised 2 PhD students and over 20 undergraduates, guiding them from circuit design basics to clinical trial execution. Two of my mentees successfully transitioned to PhD programs at Duke University and UNC Chapel Hill under my guidance.

Commitment to Health Equity: Passionate about equitable healthcare, I led a team of five to win the **MIT Hacking Racism in Healthcare Challenge** (400+ participants), developing a rapid-response solution for maternal health disparities in 48 hours. Additionally, I volunteered as a mentor for **STEAM for Vietnam**, teaching coding to nearly 7,000 children to democratize access to technical education.

Conclusion: Readiness for the Biswas Fellowship

My career has served as a rigorous training ground, equipping me with dual expertise to navigate both cutting-edge research and commercial product development. I possess the **academic depth** to design novel bio-interfaces (PhD work) and the **industrial breadth** to ensure they are robust, safe, and scalable (Vena Vitals). The Biswas Fellowship represents the natural evolution of my work. Having spent years perfecting how we **sense** the body (Input), I am now prepared to close the loop and explore how we can **modulate** it (Output). The proposed "Nasal Autonomic Interface" is the synthesis of my expertise in low-noise hardware, edge-computing, and clinical translation, poised to deliver a new class of accessible therapeutics for mental and respiratory health.